

Stacking & Blending — Simple Notes

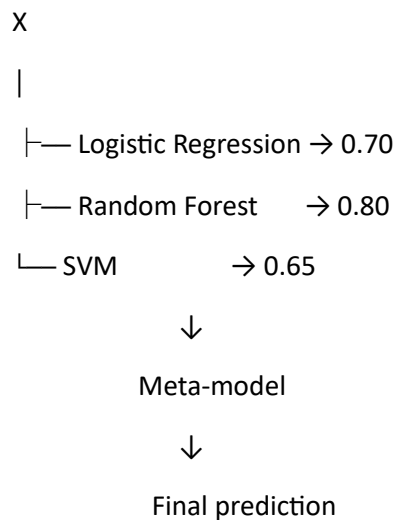
Both are **ensemble learning techniques**.

The main idea is:

Instead of trusting one model, use predictions from multiple models and let another model combine them.

1. What is Stacking?

Suppose we have 3 base models:



The **base models** make predictions.

The **meta-model** learns how to combine those predictions.

Example

Suppose:

Logistic Regression → 0.70

Random Forest → 0.80

SVM → 0.65

These become new features:

$$X_{meta} = [0.70, 0.80, 0.65]$$

The meta-model uses these to make the final prediction.

Key difference from Voting

Voting:

We manually combine predictions.

Stacking:

A model **learns how to combine** predictions.

2. The Problem: Overfitting

Suppose we train a Random Forest on training data.

Then we ask that same Random Forest to predict the **same training data**.

It may produce extremely good predictions because it has already seen those examples.

Then:

Training data



Base model



Predictions



Meta-model

The meta-model may learn from overly optimistic predictions.

➡ **High risk of overfitting.**

3. Blending — Holdout Method

Blending solves this using a separate validation set.

Suppose:

Dataset



Train 70%

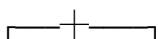
Validation 15%

Test 15%

Step 1

Train base models only on **training data**.

Training data



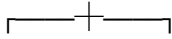
LR RF SVM

Step 2

Use those models to predict the **validation data**.

Validation data

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LR RF SVM

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.7 .8 .65

Step 3

Use those predictions to train the meta-model.

[.7, .8, .65]

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Meta-model

Step 4

Finally evaluate the complete ensemble on the **test set**.

4. Stacking — K-Fold Method

Instead of sacrificing a fixed validation portion, stacking commonly uses **K-fold cross-validation** to generate predictions for the meta-model.

Suppose:

$$K = 5$$

Training data:

Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5

Round 1

Train on:

2 3 4 5

Predict:

Fold 1

Round 2

Train on:

1 3 4 5

Predict:

Fold 2

And continue.

Eventually every training example receives a prediction from a model that **didn't train on that example**.

These are called **out-of-fold (OOF) predictions**.

OOF predictions

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Meta-model training

This is the key idea behind proper stacking.

5. Blending vs Stacking

Blending

Holdout validation set

Simpler

Less training computation

Uses fixed validation data

Can lose some training data for base models

Stacking

K-fold cross-validation

More sophisticated

More computation

Uses all training samples for OOF predictions

Better use of training data

Easy memory:

Blending = Holdout

Stacking = K-Fold + OOF predictions

6. Multi-Layer Stacking

You can have multiple layers.

Input

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Layer 1

└─ Model A

└─ Model B

└─ Model C

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Layer 2

└─ Model D

└─ Model E

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Final Model

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Prediction

The output of one layer becomes input to the next layer.

This can create a very complex ensemble.
